



Effects of *Atractylodes macrocephala* Koidzumi rhizome on 3T3-L1 adipogenesis and an animal model of obesity

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ARTICLE INFO

Article history:

Received 4 March 2011

Received in revised form 6 May 2011

Accepted 30 May 2011

Available online 6 June 2011

Keywords:

Atractylodes macrocephala Koidzumi

Obesity

3T3-L1 adipocyte

Oil Red O staining

Phospho-Akt

Perilipin

ABSTRACT

Ethnopharmacological relevance: *Atractylodes macrocephala* Koidzumi (AMK) is an herbal medicine traditionally used for treatment of abdominal pain, gastrointestinal disease, obesity, and related complications.

Aim of the study: We investigated the effects and molecular mechanism of AMK rhizome water extract on 3T3-L1 adipogenesis and an animal model of obesity.

Materials and methods: To study the effect of AMK on adipogenesis *in vitro*, differentiating 3T3-L1 cells were treated every two days with AMK at various concentrations (1–25 µg/ml) for eight days. Oil Red O staining was performed to determine the lipid accumulation in 3T3-L1 cells. To elucidate the inhibitory mechanism of AMK on adipogenesis, phosphorylation levels of Akt and expression of perilipin, were analyzed by Western blotting. AMK was administered orally to high fat diet (HFD)-induced obese rats to confirm its effect *in vivo*.

Results: AMK inhibited 3T3-L1 adipocyte differentiation in a dose-dependent manner without cellular toxicity. Phospho-Akt expression was highly decreased by AMK treatment, whereas there was no significant change in perilipin expression. AMK administration significantly reduced the body weight of rats fed a HFD. Plasma triglyceride levels were significantly lower in the AMK-treated HFD group than those in the HFD control group or normal diet (ND) group, although serum total, HDL- and LDL-cholesterol levels did not differ between the groups.

Conclusion: These results demonstrate an inhibitory effect of AMK on adipogenesis through reduction of an adipogenic factor, phospho-Akt. AMK had a beneficial effect, reducing body weight gain in a HFD-induced animal model of obesity.

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1. Introduction

Obesity has become one of the most common metabolic diseases throughout the world. More than 1 billion adults are classified as overweight. Obesity and being overweight are known to lead to adverse metabolic effects on insulin resistance, blood pressure, and cholesterol and triglyceride concentrations in plasma. These unfavorable consequences are caused by increased adipocyte numbers and adipocyte mass, which results from a massive adipocyte

differentiation process which generates mature adipocytes from preadipocytes. Excess fat is accumulated in adipocytes as triglyceride and massive amounts of lipids can be released from adipocytes, resulting in elevated triglyceride content in plasma and tissues including muscle and liver, which can lead to physiologic dysfunction in tissues (Tang et al., 1999; Frühbeck et al., 2001).

Adipocyte differentiation is a complex process that involves expression of several adipocyte-specific genes including peroxisome proliferator-activated receptor-gamma (PPARγ), sterol regulatory element binding protein-1c (SREBP-1c), CCAAT/enhancer binding protein-alpha (C/EBPα) and fatty acid synthase (FAS), which lead to morphological changes and lipid accumulation within the cells (Christy et al., 1991; Chawla et al., 1994). It is well established that activation of the serine/threonine kinase Akt (PKB) (PI3K-Akt/PKB) pathway is the major signaling mechanism in adipocyte differentiation by which insulin and certain growth

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factors stimulate adipogenesis. Treatment with PI3K inhibitors has been shown to completely block the differentiation process of 3T3-L1 preadipocytes, indicating that PI3K is necessary for adipocyte differentiation (Tomiya et al., 1995; Magun et al., 1996; Christoffersen et al., 1998; Xia and Serrero, 1999). Moreover, Akt is a downstream effector of the PI3K pathway and its activation in 3T3-L1 cells results in differentiation into adipocytes, suggesting the involvement of Akt-mediated signaling in adipocyte differentiation (Kohn et al., 1996). On the other hand, activation of cAMP-dependent kinase (PKA) stimulates lipolysis which results from degradation of accumulated triglycerides into glycerol and nonesterified fatty acids (NEFA) in adipocytes. Lipolysis involves several lipases, lipid hydrolase, and lipid droplet-containing proteins like perilipin (Tansey et al., 2001, 2004). Perilipin in particular binds to hormone-sensitive lipase (HSL) and promotes adipocyte lipolysis via phosphorylation-dependent and independent mechanisms. Most work regarding lipolysis has showed that overexpression of perilipin A increases storage of triglycerides in fibroblasts in association with a decrease in lipolysis, and its expression is highly increased during the differentiation of preadipocytes to adipocytes (Souza et al., 1998, 2002).

Despite short-term benefits of drug treatment for obesity, medication-induced weight loss is often associated with side effects from these remedies as well as rebound weight gain when they are discontinued (Abdollahi and Afshar-Imani, 2003). Recently, many researchers have shown that natural compounds from plants like herbal medicines and their derivatives are effective in the management of obesity without significant adverse effects or mortality. For example, Ginsenoside Rh2 was successfully shown to be one of the active components in Ginseng that exerts anti-diabetic effects and prevents obesity in association with the AMPK signaling pathway in 3T3-L1 adipocytes (Hwang et al., 2007). Moreover, several animal studies demonstrated that a variety of herbal medicines such as *Zingiber officinale* (Han et al., 2005), bitter melon (*Momordica charantia*) (Huang et al., 2008), ephedra (Jeong et al., 2008), and ginseng (Kim et al., 2005) have beneficial effects leading to significant weight loss or inhibition of weight gain in high fat diet (HFD)-induced obesity models.

Atractylodes macrocephala Koidzumi (AMK) belonging to the Compositae family, is an herbal medicine traditionally used in East Asia for treatment of abdominal pain, gastrointestinal disease, obesity, and related complications (China Pharmacopoeia Commission, 2005). It has been reported that extracts from AMK have various pharmacological activities associated with anti-tumor (Mori et al., 1989; Tsuneki et al., 2005; Kimura, 2006), anti-lipid-peroxidation (Kiso et al., 1985), antiulcer (Kubo et al., 1983; Nogami et al., 1986; Matsuda et al., 1991) and anti-inflammation activities (Endo et al., 1979; Li et al., 2007a,b). However, the mechanism responsible for these anti-obesity effects has not been discovered.

In this study, to evaluate the potential anti-obesity effects of AMK, 3T3-L1 preadipocytes were differentiated by inducing agents. Lipid accumulation in the cells was measured by Oil-Red O staining. Among 10 herbal medicines, AMK water extract significantly suppressed lipid accumulation with a concomitant decrease in AKT phosphorylation. In addition to this *in vitro* study, AMK was orally administered to rats fed a HFD to demonstrate its anti-obesity activities *in vivo*. Our results showed that AMK can decrease HFD-induced body weight gain as well as triglyceride concentration in blood.

2. Materials and methods

2.1. Preparation of herbal extracts

Ten herbs including *Atractylodes macrocephala* Koidzumi (AMK) were purchased from the Oriental Medicine of Kyung Hee Medi-

cal Center, Kyung Hee University (Seoul, Korea), and the voucher specimen was deposited at the laboratory of Medical Science, Graduate school of East-West Medical Science, Kyung Hee University. The dried herbs were soaked in 10 volumes of cold distilled water for overnight and then were extracted after being boiled for 3 h at 100 °C. The aqueous extracts were filtered through a distilled gauze and concentrated using an evaporator. The extracted herbal medicines were dried with a freeze dryer and stored at –20 °C until the experiment was performed. The concentration used in the experiment was based on the dry weight of the extract.

2.2. 3T3-L1 cell culture and stimulation

Murine 3T3-L1 preadipocytes (ATCC, Manassas, VA, USA) were maintained in Dulbecco's Modified Eagle's Medium (DMEM; Gibco/BRL, Grand Island, NY, USA) supplemented with 10% (v/v) BCS (JBI, Daegu, Korea) and 1% (v/v) antibiotics (Gibco/BRL) in a humidified atmosphere of 95% air and 5% CO₂ at 37 °C. To differentiate preadipocytes into adipocytes, cells were seeded at a density of 2.4×10^5 cells/cm². After four days of growth the media was changed to adipocyte-induction media I (DMEM containing 10% [v/v] fetal bovine serum [FBS; JBI], 1% [v/v] antibiotics [Gibco/BRL], 0.5 mM isobutylmethylxanthine, 1 μM dexamethasone, and 10 μg/ml insulin). After three days, cells were changed to adipocyte-induction media II (DMEM containing 10% [v/v] FBS, 1% [v/v] antibiotics and 10 μg/ml insulin) for five additional days. Finally on day eight, 3T3-L1 preadipocytes showed the adipocyte phenotype with accumulation of lipid droplets. Media was changed every two days. Appearance was recorded by microscopic using 200× magnification with an Axiovert S 100 (Carl Zeiss) equipped with an AxioCam.

2.3. Treatment of herb extracts on 3T3-L1 cells

Herb extracts were dissolved in adipocyte-induction media and filtered through 0.2 μm-pore syringe filters. To investigate the effects of herb extracts on adipogenesis, differentiating 3T3-L1 cells were treated every two days with an herb in adipocyte-induction media at various concentrations (1–25 μg/ml) for eight days.

2.4. Oil Red O staining

The 3T3-L1 adipocytes were washed with phosphate-buffered saline (PBS, pH 7.4) and fixed with 4% paraformaldehyde (PFA) for 1 h. After washing with PBS, the cells were stained with Oil Red O solution (0.2% [w/v] in isopropanol) for 1.5 h at room temperature. The cells were washed in distilled water three times and Oil Red O dye in lipid droplets was eluted into isopropanol. Finally, absorbance was measured at 490 nm using an enzyme-linked immunosorbent assay (ELISA) reader (Emax, USA) ($n = 4$).

2.5. Cell viability assay

Cell viability was determined by MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) assay (Sigma Aldrich, St. Louis, MO, USA). At the end of the experimental period (day 8), 100 μl of MTT solution (5 mg/ml) was added to differentiated 3T3-L1 cells in a 24-well plate. After the well was incubated for 2 h at 37 °C, the medium was removed, and the synthesized formazan crystals were dissolved in dimethyl sulfoxide (DMSO) (Sigma). Finally, absorbance was measured at 570 nm using an ELISA reader (Emax, USA) ($n = 4$). Data was calculated as a percentage of MTT compare to control cells.

Table 1
Composition of experimental diets.

Ingredient (%)	Normal diet (ND)	High fat diet (HFD)
Casein, 80 Mesh	19.0	25.8
L-cystine	0.3	0.4
Corn starch	29.9	0.0
Maltodextrin 10	3.3	16.2
Sucrose	33.2	8.9
Cellulose, BW200	4.7	6.5
Soybean oil	2.4	3.2
Lard	1.9	31.7
Mineral mix S10026	0.9	1.3
Dicalcium phosphate	1.2	1.7
Calcium carbonate	0.5	0.7
Potassium citrate	1.6	2.1
Vitamin mix V10001	0.9	1.3
Choline bitartrate	0.2	0.3
Percent energy		
Protein	20.0	20.0
Carbohydrate	70.0	20.0
Lipid	10.0	60.0

2.6. Western blot analysis

Cells were washed three times with cold PBS and scraped in lysis buffer (20 mM Tris–HCl [pH 7.4], 150 mM NaCl, 10% glycerol, 2% Nonidet P-40, 1 mM EDTA, 20 mM sodium fluoride, 30 mM sodium pyrophosphate, 0.2% sodium dodecyl sulfate (SDS), 0.5% sodium deoxycholate, 1 mM phenylmethylsulfonyl fluoride (PMSF), 1 mM dithiothreitol (DTT), 1 mM sodium vanadate). The cell lysate was incubated on ice while vortexing for 15 min and centrifuged for 20 min at 12,000 × g. Protein quantification of each supernatant was measured by Bradford assay (Ref). Each protein extract (5, 10, or 20 µg) was separated on 10% SDS-polyacrylamide gel onto 0.2-µm polyvinylidene difluoride (PVDF) membranes (Bio-Rad, Hercules, CA, USA) using a semi-dry transfer apparatus (Trans-Blot SD; Bio-Rad). The membranes were then incubated with primary antibody diluted (Akt-p [1:1000], Akt [1:2000], perilipin [1:2000] actin [1:1000]) in blocking solution at 4 °C overnight. Membranes were probed with horseradish peroxidase-labeled secondary antibody (anti-goat IgG or anti-rabbit IgG) for 1 h at room temperature. The immunoreactive bands were detected by chemiluminescence using West-one™ (iNtRON Biotechnology). All antibodies were purchased from Santa Cruz Biotechnology and Abcam.

2.7. Image analysis

All image analysis was performed with ImageMaster 2D™ Elite software, version 3.1 (Amersham Pharmacia Biotechnology). The signal intensity of each band was analyzed by lowest background subtraction and normalized by band intensity of Akt (for Akt-p) and actin (for perilipin).

2.8. In vivo anti-obesity activity

Six-week-old male Sprague-Dawley rats weighing 170–190 g (Samtaco, Seoul, Korea) were housed four per cage in a laminar air-flow room maintained at a temperature of 22 ± 1 °C and a relative humidity of 55 ± 1%. After the animals were given a standard laboratory diet for one week, they were randomly divided into a normal diet group (ND) (n = 6), high-fat diet group (HFD) (n = 6), and high-fat diet + AMK group (HFD + AMK) (n = 6). Table 1 shows the HFD composition. Rats in the HFD group were orally administered saline once a day while rats in the HFD + AMK group were orally administered 0.5 g AMK/kg body weight in saline. All experimental rats were allowed free access to the diet and water during the experimental period. The research was conducted in accordance with the internationally accepted principles for laboratory animal use and

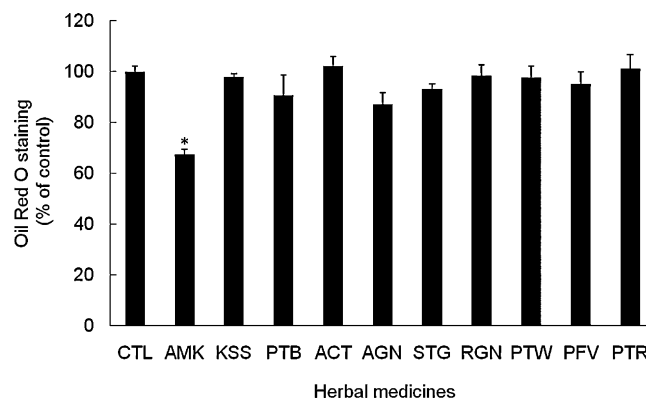


Fig. 1. Effect of various herbal medicines on cellular lipid droplets. Differentiating 3T3-L1 cells were treated every 2 days with 10 µg/ml of herbal medicines (from day 0 to day 8) in adipocyte-induction media. To determine lipid content accumulation, the cells were stained with Oil Red O at day 8 and optical density detected at 490 nm. Data was calculated as a percentage of Oil Red O stained lipid in controls. Results represent the mean ± SEM of four independent experiments (*p < 0.001). CTL (control), AMK (*Atractylodes macrocephala* Koidzumi), KSS (*Kochia scoparia* Schrader), PTB (*Pinellia ternata* Breitenbach), ACT (*Artemisia capillaris* Thunberg), AGN (*Angelica gigas* Nakai), STG (*Sophora tonkinensis* Gagnep), RGL (*Rehmannia glutinosa* Liboschitz var.), PTW (*Polygala tenuifolia* Willdenow), PFV (*Perilla frutescens* var. *acuta*), PTR (*Poncirus trifoliata* Rafinesqu).

care as found in the US guidelines (NIH publication #85-23, revised in 1985). Food intake and body weights were recorded daily for six weeks. The rats were euthanized by anesthetic overdose and blood was taken for triglyceride, total cholesterol, and HDL cholesterol analysis (ARC Laboratory, South Korea).

2.9. Statistical analysis

Data are presented as mean ± SEM. Analysis was performed by ANOVA with Bonferroni's test for multiple comparisons, and by *t*-test, using SPSS 11.0 software. A *p*-value of <0.05 was considered statistically significant.

3. Results

3.1. Anti-adipogenic effects of 10 herbal medicines on 3T3-L1 adipocytes

To evaluate the anti-adipogenic effects of 10 herbal medicines, post-confluent 3T3-L1 preadipocytes were treated with 10 µg/ml of an herbal medicine every two days while the preadipocytes differentiated into adipocytes. Morphological changes were observed due to accumulation of lipids in the preadipocytes. Oil Red O staining revealed that the lipid accumulation in AMK-treated cells was significantly lower than the lipid accumulation in control cells and other herb-treated cells. Of the 10 herbal treatments, AMK-treated cells showed a 33% inhibitory effect on adipogenesis of 3T3-L1 adipocytes (Fig. 1). Therefore, AMK was subjected to additional analysis to better understand its anti-adipogenic effect on 3T3-L1 adipocytes.

3.2. AMK inhibits 3T3-L1 adipocyte differentiation

To examine the anti-adipogenic effect of AMK on 3T3-L1 adipocyte differentiation, post-confluent 3T3-L1 preadipocytes were maintained in adipocyte-induction media and exposed to various doses of AMK (1, 5, 10, 25 µg/ml). Subsequently, intracellular lipid drops were measured by Oil Red O staining. As shown in Fig. 2(B), AMK effectively inhibited adipocyte differentiation in a dose-dependent manner in 3T3-L1 adipocytes. We next examined intracellular toxicity by MTT assay. Fig. 2(C) shows that viability of

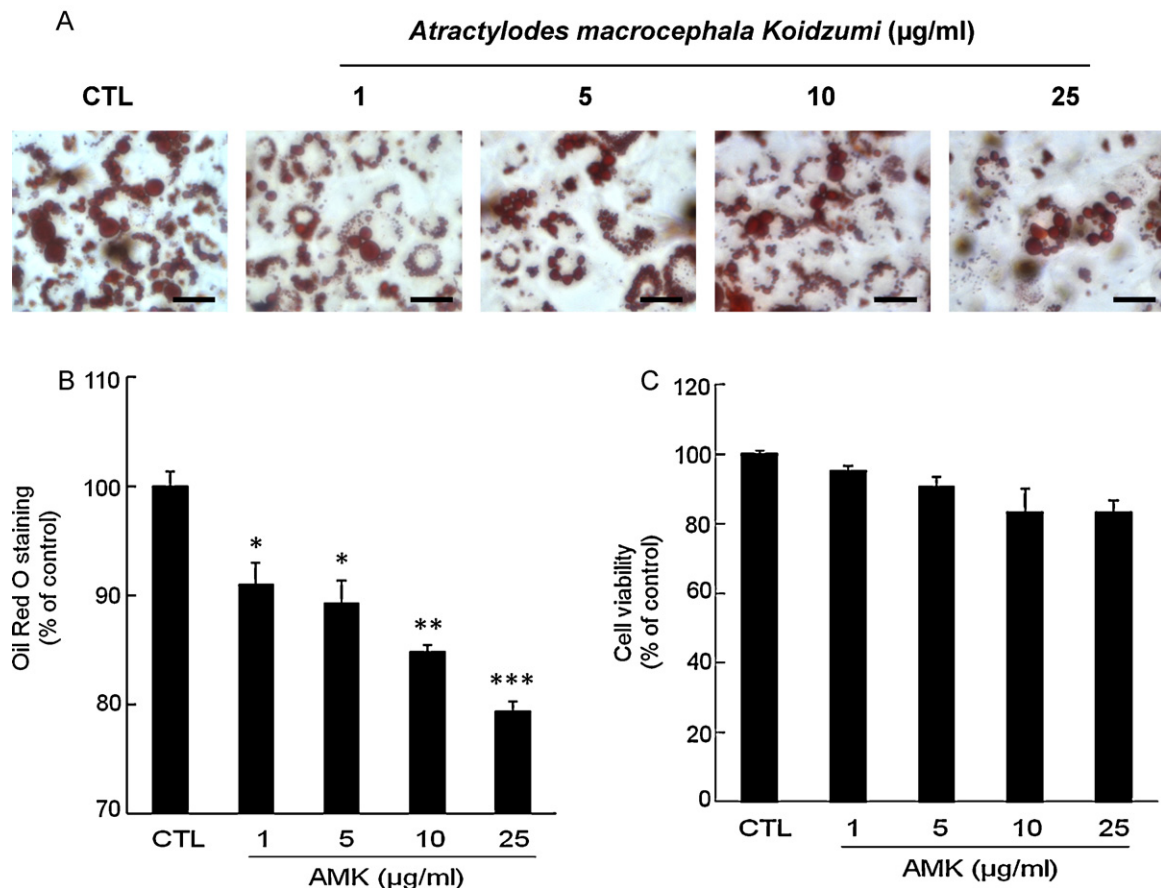


Fig. 2. Effect of AMK on cellular lipid droplets and cell viability of 3T3-L1 adipocytes. Differentiating 3T3-L1 cells were treated every 2 days with AMK at various concentrations (from 1 to 25 µg/ml) for 8 days in adipocyte-induction media. Intracellular lipids were stained with Oil Red O (A) (scale bar; 20 µm). To determine the accumulation of lipid content, Oil Red O dye was dissolved in isopropanol and optical density detected at 490 nm (B). Data was calculated as a percentage of Oil Red O stained lipid in controls. Results represent the mean ± SEM of four independent experiments (* $p < 0.01$, ** $p < 0.001$ and *** $p < 0.0001$ vs. control). (C) 3T3-L1 preadipocytes were maintained with adipocyte-induction media for 8 days and treated with AMK every 2 days (see Section 2). Cell viability was calculated as a percentage of MTT metabolisms in controls. Results represent the mean ± SEM of four independent experiments.

AMK-treated cells was not significantly affected at concentrations up to 25 µg/ml. The cell viability result indicates that the reduction of lipid accumulation in AMK-treated cells was not due to cellular toxicity. Therefore, we concluded that the reduction in Oil Red O staining is mainly a result of an anti-adipogenic effect of AMK leading to an inhibition in 3T3-L1 adipocyte differentiation.

3.3. Effects of AMK on phospho-Akt and perilipin in 3T3-L1 adipocytes

Serine/threonine kinase Akt has been demonstrated to be activated by insulin and is a downstream target of phosphatidylinositol 3-kinase (PI3Ks) in adipocyte differentiation (Kohn et al., 1996). To reveal the molecular mechanisms underlying the AMK-induced anti-adipogenic effect, we first examined the effect of AMK on the level of phosphorylated Akt (phospho-Akt) by Western blot from 3T3-L1 cell lysates treated with AMK at various concentrations (1, 5, 10, 25 µg/ml) during the time preadipocytes differentiated into adipocytes. Treatment with AMK dose dependently decreased the level of phospho-Akt compared to the control cells (Fig. 3). Since the inhibition of lipid accumulation in adipocyte differentiation can also be affected by lipolysis along with decreased expression of the lipid droplet-associated protein, perilipin (Souza et al., 1998, 2002), we next determined whether perilipin was involved in the AMK-induced anti-adipogenic effect on 3T3-L1 preadipocyte differentiation. As shown in Fig. 3, Western blots from 3T3-L1 cell lysates

treated with AMK at various concentrations (1, 5, 10, 25 µg/ml) revealed that perilipin protein expression was not significantly changed by AMK treatment. Taken together, these results suggest that the Akt signaling pathway plays an important role in the anti-adipogenic effect induced by AMK treatment on 3T3-L1 preadipocyte differentiation and AMK has no effect on the activation of lipolysis via perilipin expression.

3.4. Effect of AMK on HFD-induced obese rats

To investigate the anti-adipogenic properties of AMK *in vivo*, we recorded body weights after oral administration of AMK 0.5 g/kg per day to HFD-induced obese rats for six weeks. Table 2 shows

Table 2
Weight gain and food intake in ND, HFD, and HFD-AMK groups for six weeks.

	ND ^a	HFD ^b	HFD-AMK ^c
Body weight (g)			
Initial	183.6 ± 18.40	183.7 ± 2.993	181.7 ± 0.952
Final	454.9 ± 16.14	495.0 ± 27.95	483.4 ± 8.813*
Food intake (g)	177.8 ± 3.130	143.1 ± 20.03	141.1 ± 1.882

Data represent the mean ± SEM, * $p < 0.05$, when compared to the HFD group ($n = 6$ per group).

^a Normal diet fed group.

^b High-fat diet fed group.

^c High-fat diet + AMK (0.5 g/kg body weight/daily).

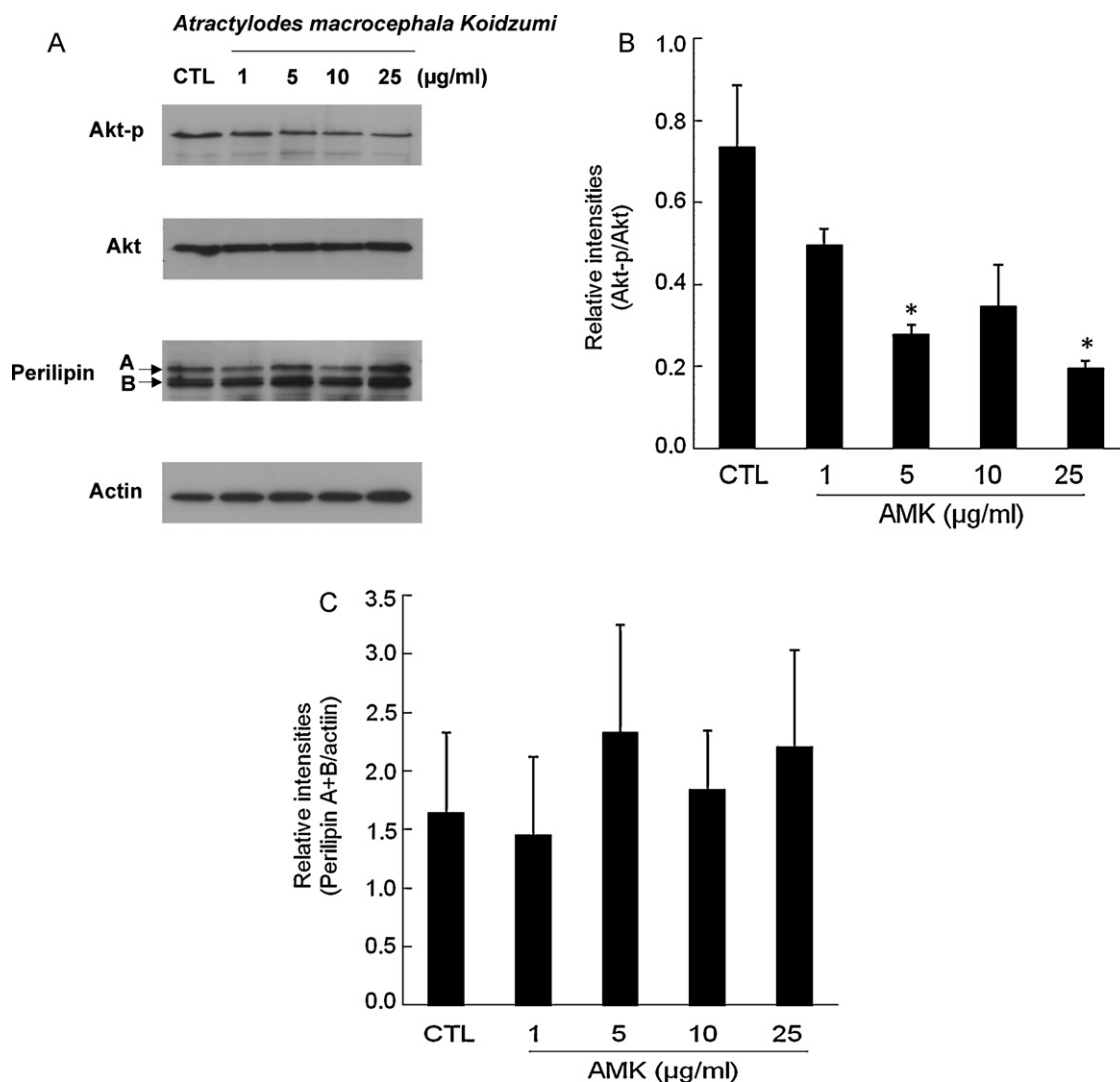


Fig. 3. Effect of AMK on Akt-p and Perilipin A/B expression. Differentiating 3T3-L1 cells were treated with AMK at various concentrations (from 1 to 25 µg/ml) for 8 days in adipocyte-induction media. Western blot analysis for key adipogenic factors including Akt-p and perilipin was conducted. (A) Expression levels of Akt-p (20 µg) and perilipin (10 µg) were normalized by Akt (20 µg) and actin (5 µg) using ImageMaster™ 2D Elite software and quantified as relative intensities (B) and (C). Results represent mean ± SEM of three independent experiments (* $p < 0.05$ vs. control).

that the HFD increased body weight significantly compared to the normal diet (ND) over the six week period. Moreover, final body weight was significantly lower in the HFD + AMK group compared to the HFD group ($p < 0.05$) after six weeks without affecting food intake (Table 2).

The concentration of plasma triglycerides was significantly influenced by the HFD. Fig. 4(A) shows that plasma triglyceride concentration in the HFD group increased remarkably from 111 to 162 mg/dL compared to the ND group. HFD rats administered AMK had lower plasma triglyceride concentrations than rats fed only the HFD; however, the HFD did not affect the total concentrations of HDL and LDL in plasma (Fig. 4(B)). AMK slightly changed the concentration of HDL and LDL in plasma but there was no significant difference between the HFD group and the HFD + AMK group. Although there were no significant difference observed in total cholesterol, HDL, and LDL, these results indicate that oral administration of AMK can decrease HFD-induced body weight gain as well as triglyceride concentration in the blood.

4. Discussion

Obesity is associated with many pathological disorders, including diabetes (Sartipy and Loskutoff, 2003), hypertension (Pi-Sunyer,

2002), cancer (Lagra et al., 2004) and atherosclerosis (Wofford et al., 1999). Thus the search for possible compounds to treat obesity has intensified. Despite the fact that whole extracts and active compounds from plants have possible anti-obesity pharmacological activities, the efficacy and action mechanisms of relatively few herbal medicines have been demonstrated compared to Western drugs.

AMK has traditionally been known as a treatment for obesity and related complications (Chemical Industry Press, 2005); however, the mechanisms involved in its anti-obesity effects using models of adipogenesis *in vitro* and *in vivo* have not yet been investigated. In this study it was found that lipid accumulation inside differentiated 3T3-L1 adipocytes was significantly decreased by treatment with the rhizome of AMK water extract.

AMK contains many bioactive compounds including atractylenolide I, atractylenolide II, and atractylon. These are reported to have inhibitory effects on inflammation (Dong et al., 2008). Atractylenolide I and atractylenolide II have been demonstrated to induce cell differentiation and suppress cell migration by inhibiting Ras/ERK MAPK and PI3/Akt pathways in B16 melanoma cells (Ye et al., 2010). The PI3K/Akt pathway is also known as the major signal transduction pathway in adipogenesis activated

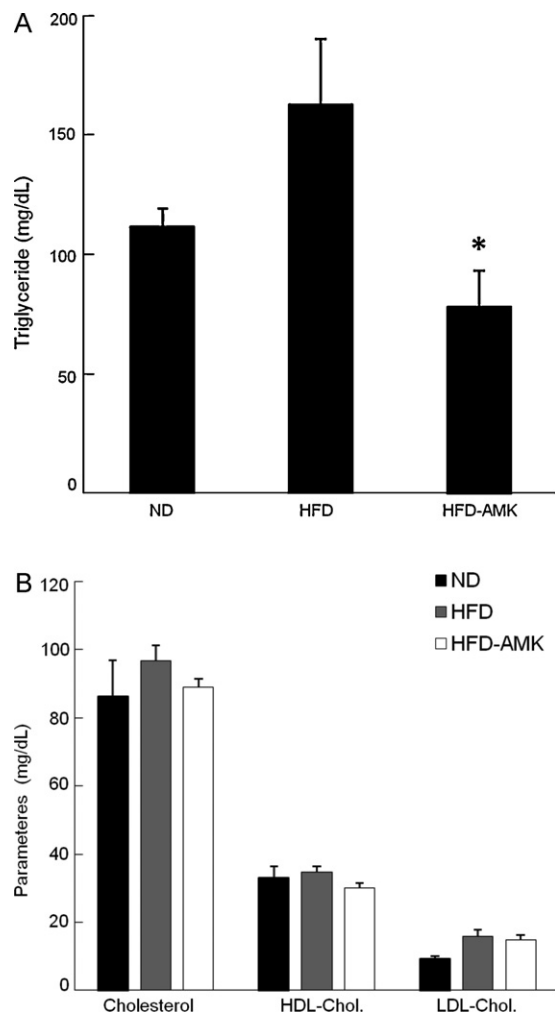


Fig. 4. Effect of AMK on lipid profiles in normal diet (ND), high fat diet (HFD), and HFD + AMK-fed rats after six weeks. (A) Effect of AMK on triglyceride level and (B) total cholesterol, HDL-C, and LDL-C in HFD rats. Results represent mean $\pm$ SEM (* p < 0.05 vs. control, n = 6 per group).

by insulin/IGF-1 signaling. In our study, Western blot results revealed that treatment with AMK at various concentrations (1, 5, 10, 25 μ g/ml) dose-dependently decreased the level of phospho-Akt (Fig. 3). Although we did not measure atractylenolide I and atractylenolide II concentrations in the AMK water extract, it is speculated that atractylenolide I and atractylenolide II may mainly affect the reduced accumulation of triglyceride formation by inhibition of the PI3/Akt pathway during differentiation of 3T3-L1 preadipocytes into adipocytes.

In this study, we also used a HFD-induced obesity rat model to verify the anti-obesity effects of AMK water extract *in vivo*. Body weights of HFD-induced obese rats were monitored after daily oral administration of AMK 0.5 g/kg for six weeks. An interesting finding in the *in vivo* study was that AMK caused decrease weight gain in HFD rats along with decreased triglyceride concentrations in the blood (Fig. 4(A)), but it did not affect food intake (Table 2), thus raising the possibility of an anti-adipogenic mechanism by which AMK affects adipogenesis and energy expenditure. Although we did not examine the effect of AMK on inhibition of the PI3K pathway in adipocytes from HFD-induced obese rats, this hypothesis is supported by our *in vitro* study which demonstrated that AMK decreased lipid accumulation by inhibition of the PI3K pathway in 3T3-L1 preadipocytes differentiated into adipocytes. It is also possible that energy expenditure may be increased in rats orally

administered AMK; however, this possibility should be confirmed by further investigation.

A number of clinical studies considering the anti-obesity effects of herbal medicines in human have shown a significant decrease in body fat and appetite. Most of the studies reported no significant diverse effects compared to controls except in studies with supplements containing ephedra and caffeine (Hasani-Ranjbar et al., 2009) which caused minor adverse effects such as headache, dry mouth, and nervousness. It is worth noting that despite showing significant decreases in body fat and energy intake in humans treated with herbal medicines, long-term studies investigating the effects of herbal medicines on body weight and lipid metabolism as well as identifying their pharmacokinetics and optimal doses must be performed to avoid the possibility of adverse effects, before they can be touted as effective alternative therapies for treatment of obesity.

In the present study, we evaluated the anti-obesity effects of AMK on adipocyte differentiation and associated mechanisms in mouse 3T3-L1 cells and confirmed our findings in an obese animal model fed a HFD. AMK water extract greatly decrease lipid accumulation in 3T3-L1 adipocytes by inhibition of the Akt/PI3K pathway. Moreover, administration of AMK effectively suppressed body weight gain and decreased plasma triglycerides in rats fed a HFD. These results suggest that the anti-obesity effect of AMK results from a decrease in adipogenesis and that AMK has a beneficial effect, reducing body weight gain in an experimental animal model.

Acknowledgement

This study was funded by the program of the Kyung Hee University for the young medical researcher in 20071480.

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